

High peak and high average radiofrequency power transmit/receive switch for thermal magnetic resonance

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Purpose: To study the role of temperature in biological systems, diagnostic contrasts and thermal therapies, RF pulses for MR spin excitation can be deliberately used to apply a thermal stimulus. This application requires dedicated transmit/receive (Tx/Rx) switches that support high peak powers for MRI and high average powers for RF heating. To meet this goal, we propose a high-performance Tx/Rx switch based on positive-intrinsic-negative diodes and quarter-wavelength ($\lambda/4$) stubs.

Methods: The $\lambda/4$ stubs in the proposed Tx/Rx switch design route the transmitted RF signal directly to the RF coil/antenna without passing through any electronic components (e.g., positive-intrinsic-negative diodes). Bench measurements, MRI, MR thermometry, and RF heating experiments were performed at $f = 297$ MHz ($B_0 = 7$ T) to examine the characteristics and applicability of the switch.

Results: The proposed design provided an isolation of -35.7 dB/ -41.5 dB during transmission/reception. The insertion loss was -0.41 dB/ -0.27 dB during transmission/reception. The switch supports high peak (3.9 kW) and high average (120 W) RF powers for MRI and RF heating at $f = 297$ MHz. High-resolution MRI of the wrist yielded image quality competitive with that obtained with a conventional Tx/Rx switch. Radiofrequency heating in phantom monitored by MR thermometry demonstrated the switch applicability for thermal modulation. Upon these findings, thermally activated release of a model drug attached to thermoresponsive polymers was demonstrated.

Conclusion: The high-power Tx/Rx switch enables thermal MR applications at 7 T, contributing to the study of the role of temperature in biological systems and diseases. All design files of the switch will be made available open source at www.opensourcemedicine.org.

KEYWORDS

open source hardware, RF induced heating, thermal magnetic resonance, transmit receive switch, ultrahigh field MR

1 | INTRODUCTION

Temperature is a physical parameter with diverse biological implications and intense clinical interest. Controlled increase of temperature has been used in thermal therapies. Mild hyperthermia performed at temperatures of $T = 41\text{--}45^\circ\text{C}$ is a potent sensitizer for chemotherapies and radiotherapies.^{1–5} Smart drug carriers that release their load after a temperature stimulus are used to deliver chemotherapeutics specifically to the target site, increasing the therapeutic dose and minimizing systemic toxicity.^{6,7} Other (bio)molecules have been loaded to thermoresponsive carriers for delivery and controlled release, such as MRI contrast agents,^{8,9} genes,^{10,11} and peptides/proteins.¹²

Magnetic resonance imaging is a valuable tool for diagnosis, treatment planning, and treatment control of thermal interventions, including the spectrum of diagnostic imaging and noninvasive temperature monitoring.^{13–16} It has been demonstrated that RF pulses tailored for spin excitation can also be used for the application of a thermal stimulus.¹⁷ This approach, denoted as thermal MR,^{18–20} uses the pulsed power amplifier of the MR system together with dedicated RF antenna arrays to generate RF-induced focal thermal modulation.

Performing MRI and RF heating simultaneously requires high peak power (>1 kW) and high average power (>100 W per channel), and support of the RF antennae as well as other RF hardware components. Although RF antennae have already been developed for thermal MR,^{17–20} high peak and high average RF power transmit/receive (Tx/Rx) switches that switch between high power (kW) transmission to low power (μW) reception of the MR signal are still lacking.

Magnetic resonance Tx/Rx switches deploy positive-intrinsic-negative (PIN) diodes as active switching units, as they offer very low resistance in forward biased state, high isolation in the reverse bias state, fast switching time, and high peak power handling.²¹ The PIN diodes have been incorporated in various designs for different purposes: high isolation,²² ultrashort switching time,²³ double-tuned nuclei in linear/quadrature mode,²⁴ and miniaturized Tx/Rx switches.²⁵ Although all of these designs are capable of handling high peak power for MRI, their average power handling (e.g., at high duty cycles) is limited typically by electronic components (e.g., PIN diodes) placed in the transmission path. Thus, the overall applicable average power is limited severely to avoid excessive heat accumulation in these components.

In this work, we propose a high peak and high average power Tx/Rx switch design that supports MRI and high-power RF heating in thermal MR applications at 297 MHz ($B_0 = 7$ T). The proposed switch was benchmarked against a conventional Tx/Rx switch. Diagnostic in vivo MRI, RF

heating, and temperature-induced release of a model drug from a thermoresponsive polymer in a phantom demonstrate the potential range of applications of the proposed Tx/Rx switch.

2 | METHODS

2.1 | Hardware design

The electrical circuit of the proposed high-power Tx/Rx switch for 7 T ($f = 297$ MHz)²⁶ is shown in Figure 1A with a photograph of the assembled switch shown in Figure 1B. In the proposed single-pole double-through design, no electronic components are placed in the direct transmission path (Figure 1C).²⁷ Consequently, excessive heat accumulation during high duty cycle high-power RF pulse transmission is avoided and higher peak and average powers can be applied.

During transmission, the PIN diodes D1 and D2 (MA4P7441F-1091T, MACOM, Lowell, MA, USA) are forward-biased ($U_{DC} = 5$ V, $I = 100$ mA). The low impedance at points 1 and 4 (Figure 1C) is transformed by the $\lambda/4$ stubs into high impedance at points 2 and 3, routing the RF pulse from the Tx port directly to the RF coil/antenna without passing through the PIN diodes.

During reception, the PIN diodes are reverse-biased ($U_{DC} = -30$ V). The high impedance at point 1 is transformed into low impedance at point 2 and again to high impedance at point 3 by the $\lambda/4$ stubs, routing the received RF signal from the RF coil/antenna into the Rx port (Figure 1D). The Rx port is connected to a low noise preamplifier (Siemens Healthineers, Erlangen, Germany).

Capacitors $C1 = C2 = 2.4$ nF and inductors $L1 = L2 = 910$ nH (3650 series, TE Connectivity, Schaffhausen, Switzerland) were used to block direct current (DC) and RF, respectively. $C1$ and $C2$ each consisted of 5 parallel 470-pF ceramic capacitors (700E, American Technical Ceramics, Huntington Station, NY, USA) and were soldered under a copper plate that was $25 \times 40 \times 2$ mm³. The PIN diodes D1 and D2 were soldered on top of these copper plates for heat dissipation (Figure 1B). The parallel circuit $L3 + C3$ tuned to 297 MHz creates DC ground for D1 and blocks RF from DC ground. The inductor $L4 = 910$ nH (3650 series, TE Connectivity) creates DC ground for D2 and acts as an RF choke (measured reactance of 1.76 k Ω at 297 MHz). The cable shield of the $\lambda/4$ stubs is connected to ground at point A (Figure 1A). At point B, a frequency-selective RF ground is provided by the series resonant circuit $L5 + C4$, with $L5$ being the self-inductance of the cable shield. For $L3 = 46$ nH, three windings (inner diameter = 4.5 mm) of 1-mm silver-coated copper wire were used. $C3 = 1\text{--}8$ pF and $C4 = 1\text{--}40$ pF are nonmagnetic trimmer capacitors (Voltronics, Salisbury, MA, USA). The $\lambda/4$ stubs ($L = 15.5$ cm) were

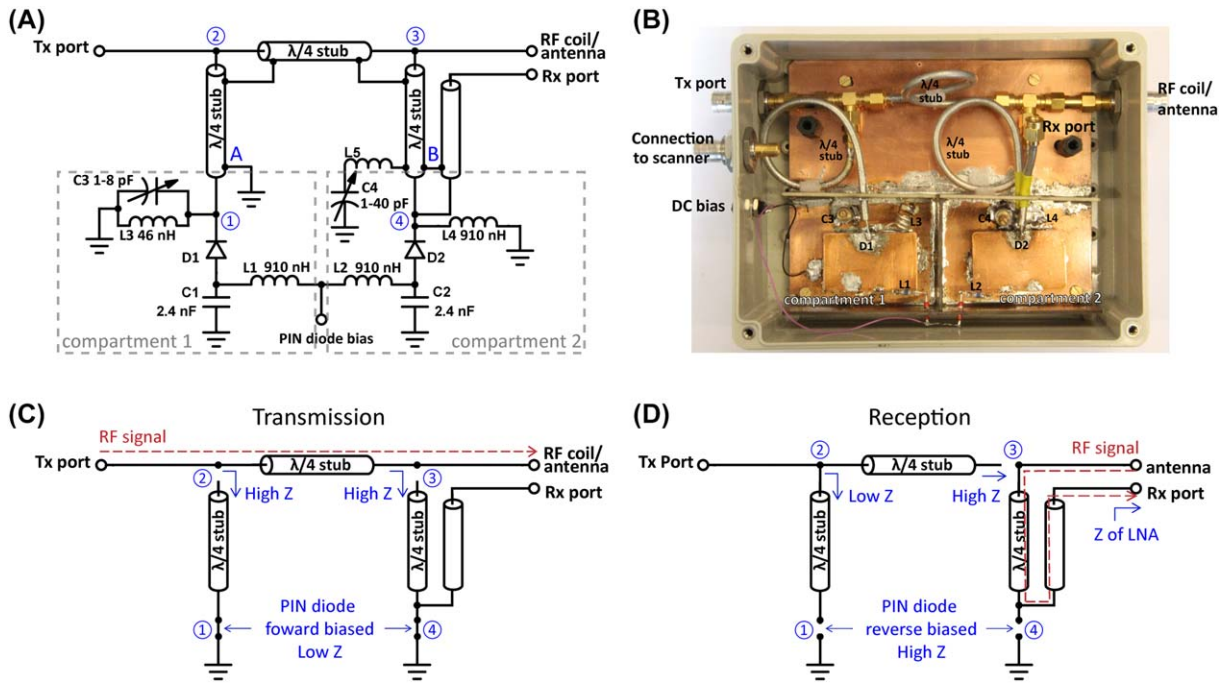


FIGURE 1 A, Electrical circuit of the proposed high-power transmit/receive (Tx/Rx) switch. Capacitors C1 and C2 and inductors L1 and L2 were used to block direct current (DC) and RF, respectively. D1 and D2 are positive-intrinsic-negative (PIN) diodes. The parallel circuit L3 + C3 (297 MHz) creates DC ground for D1 and blocks RF from DC ground. The inductor L4 creates DC ground for D2 and acts as an RF choke. The cable shield of the $\lambda/4$ stubs is connected to ground at point A. At point B, a frequency-selective RF ground is provided by the series resonant circuit L5 + C4, with L5 being the self-inductance of the cable shield. B, Photograph of the constructed Tx/Rx switch. The low-noise preamplifier (LNA) board was removed to display the full circuit. C1 and C2 (not visible) each consisted of five parallel 470 pF ceramic capacitors that were soldered under a copper plate with dimensions $25 \times 40 \times 2 \text{ mm}^3$. The PIN diodes D1 and D2 were soldered on top of these copper plates for heat dissipation. C, Equivalent circuit during the transmission mode. The PIN diodes are forward-biased; the low impedance at points 1 and 4 is transformed into high impedance at points 2 and 3 by the $\lambda/4$ stubs. In this way, the RF signal from the Tx port is routed into the RF coil/antenna without passing through the PIN diodes. D, Equivalent circuit during the receive mode. The PIN diodes are reverse-biased; the high impedance at point 1 is transformed into low impedance at point 2 and again to high impedance at point 3 by the $\lambda/4$ stubs, routing the received RF signal from the RF coil/antenna into the Rx port

assembled with semirigid coaxial cable (Sucoform 141 Cu 50 Ω , Huber+Suhner, Herisau, Switzerland) for 297 MHz. In comparison to flexible cables, semirigid cables have better dielectric properties, better shielding quality, and a more stable phase relation. The two RF branches and associated PIN diodes were placed in double-shielded compartments 1 and 2 to increase isolation (Figure 1A,B). These compartments can furthermore be characterized, calibrated, or modified individually in a modular way. This is possible due to the placement of L4 (which otherwise might not be needed, as DC ground for D2 might also be provided through L3).

The whole circuit was placed on a $110 \times 135 \times 2 \text{ mm}^3$ double-sided copper board, which also served as shielding and mounted in a $165 \times 125 \times 75 \text{ mm}^3$ casing.

The isolation and insertion loss of the Tx/Rx switch were assessed by measuring the scattering parameters using a vector network analyzer (ZVT8, Rhode & Schwarz, Memmingen, Germany).

The switching time between transmission and reception modes was measured using a digital oscilloscope (HMO3524,

Hameg Instruments, Mainhausen, Germany) and a signal generator (SMGL, Rhode & Schwarz).

2.2 | High RF power handling test

To evaluate the Tx/Rx switch under high average power conditions, $P_{\text{avg}} = 100 \text{ W}$ and $P_{\text{peak}} = 1 \text{ kW}$ (4-ms rect-pulse, TR = 40 ms, duty cycle = 10%, generated by an MR RF power amplifier – Stolberg HF Technik AG, Stolberg-Vicht, Germany) was applied to the Tx/Rx switch for 30 minutes and its temperature was monitored with an infrared camera (Ti25, Fluke, Everett, WA, USA). For comparison, a conventional 7 T Tx/Rx switch (Siemens Healthineers) was used. This switch design has PIN diodes in the transmission path (Figure 2) that limit the specified maximum power to $P_{\text{avg}} = 25 \text{ W}$.

2.3 | Magnetic resonance imaging

To demonstrate the suitability of the high-power Tx/Rx switch for MRI, we performed MRI on a human wrist in a

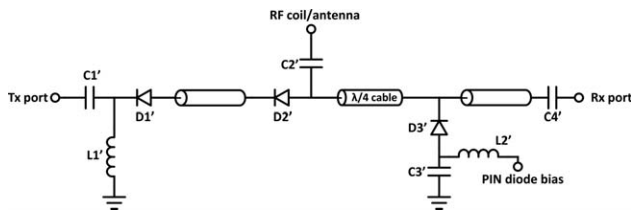


FIGURE 2 Simplified circuit schematics of the conventional Tx/Rx switch (Siemens Healthineers, Erlangen, Germany) used as a reference in this study. In the transmission mode, all PIN diodes (DH80106, Temex, Pessac, France) are forward-biased. D1' and D2' provide a low insertion loss path between the Tx port and the RF coil/antenna, whereas the receiver port is isolated by high impedance transformed by a quarter-wavelength ($\lambda/4$) cable from low impedance at D3'. In the receive mode, the PIN diodes are reverse-biased; the RF signal coming from the RF coil/antenna will be blocked from the Tx port by D1' and D2', and will be directed to the Rx port. The Rx port is connected to a low-noise preamplifier. The measured insertion loss for this switch is -0.55 dB/ -0.32 dB (transmission/reception) and the measured isolation is -35.6 dB/ -46.8 dB (transmission/reception). In this Tx/Rx switch design, all of the transmitted RF power passes through the PIN diodes D1' and D2', which is the limiting factor in determining the maximum applicable average power (< 25 W)

$B_0 = 7$ T whole-body MR system (Magnetom, Siemens Healthineers) using a bow-tie dipole antenna^{17,28} for transmission and reception of the RF signal. As a reference, the conventional Tx/Rx switch was applied in the same setup together with the same preamplifier (Siemens Healthineers). The SNR was calculated by dividing the average signal intensity of a circular region of interest drawn in the carpal bones by the standard deviation of the noise from background air. For the in vivo feasibility study, subjects were included after due approval by the local ethical committee (registration number DE/CA73/5550/09, Landesamt für Arbeitsschutz, Gesundheitsschutz und technische Sicherheit, Berlin, Germany). Informed written consent was obtained

from each volunteer prior to the study in compliance with the local institutional review board guidelines.

2.4 | Radiofrequency-induced heating

An experimental setup consisting of a rectangular agarose phantom of dimensions $90 \times 180 \times 260$ mm³ and a bow-tie dipole antenna (Figure 3A) was used to examine the suitability of the high-power Tx/Rx switch for RF heating and MR thermometry (MRTh) at 297 MHz ($B_0 = 7$ T). The agarose phantom ($\sigma = 1.03$ S/m, $\epsilon = 71.9$) was equipped with 7 cylindrical sample holders. Five were located in the RF-heated volume governed by a dipole antenna. The remaining 2 were placed outside of the RF-heated volume with $\Delta T = 0^\circ\text{C}$ (Figure 3B). The sample holders were filled with 0.1M NaCl solution ($\sigma = 1.05$ S/m, $\epsilon = 80.4$) to avoid susceptibility artifacts. PTFE tubes with 1-mm inner diameter were inserted along the long axis of the phantom at depths of 5, 15, 25, and 35 mm (Figure 3B,C) to accommodate fiber optic probes that serve as temperature reference for MRTh.

Radiofrequency heating was accomplished using the 7 T MR system amplifiers at high duty cycle (Magnetom, Siemens Healthineers). The hardware losses between the transmitter and the feeding point of the dipole antenna was -2.18 dB. This setup afforded $P_{\text{peak}} = 1$ kW and $P_{\text{avg}} = 100$ W (4-ms rect-pulse, TR = 40 ms, duty cycle = 10%) at the feeding point of the dipole antenna and $P_{\text{peak}} = 1.2$ kW and $P_{\text{avg}} = 120$ W at the high-power Tx/Rx switch. The heating paradigm consisted of 3×5 minutes of RF heating interleaved with 2D-MRTh acquisition. The 2D-MRTh was conducted using a proton resonance frequency shift method¹³ in conjunction with a dual gradient-echo technique.^{14,15} A fiber optic probe (Omniflex, Neoptix, Québec, Canada) placed 15 mm below the bow-tie dipole antenna was used as temperature reference.

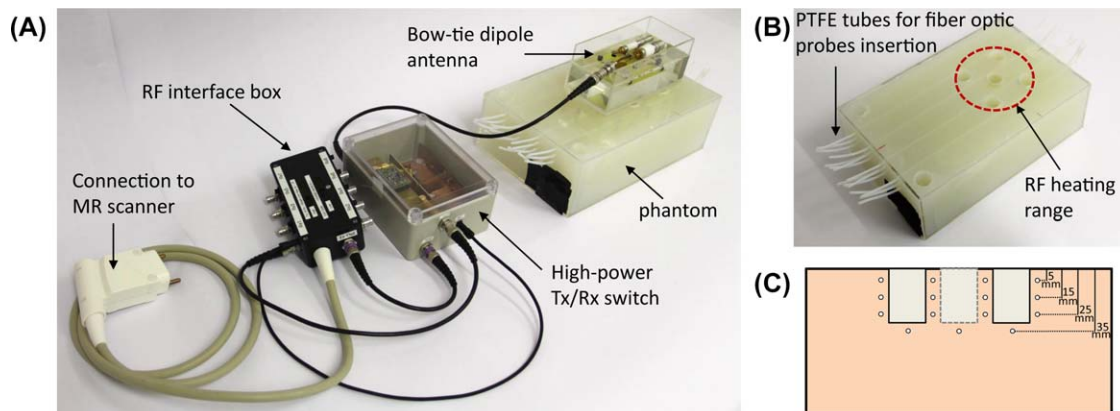


FIGURE 3 A, Photograph of the RF heating setup consisting of a rectangular agarose phantom ($90 \times 180 \times 260$ mm³, $\sigma = 1.03$ S/m, $\epsilon = 71.9$), a bow-tie dipole antenna, the high-power Tx/Rx switch, and an RF interface box. B, Photograph of the agarose phantom with seven cylindrical sample holders and 15 PTFE tubes with an inner diameter of 1 mm for fiber optic temperature probe insertion. The RF heated area is roughly marked by the red dotted circle. C, Schematic of the cross section of the phantom with three sample holders and the position of the PTFE tubes at depths of 5, 15, 25, and 35 mm

High-power switch characterization

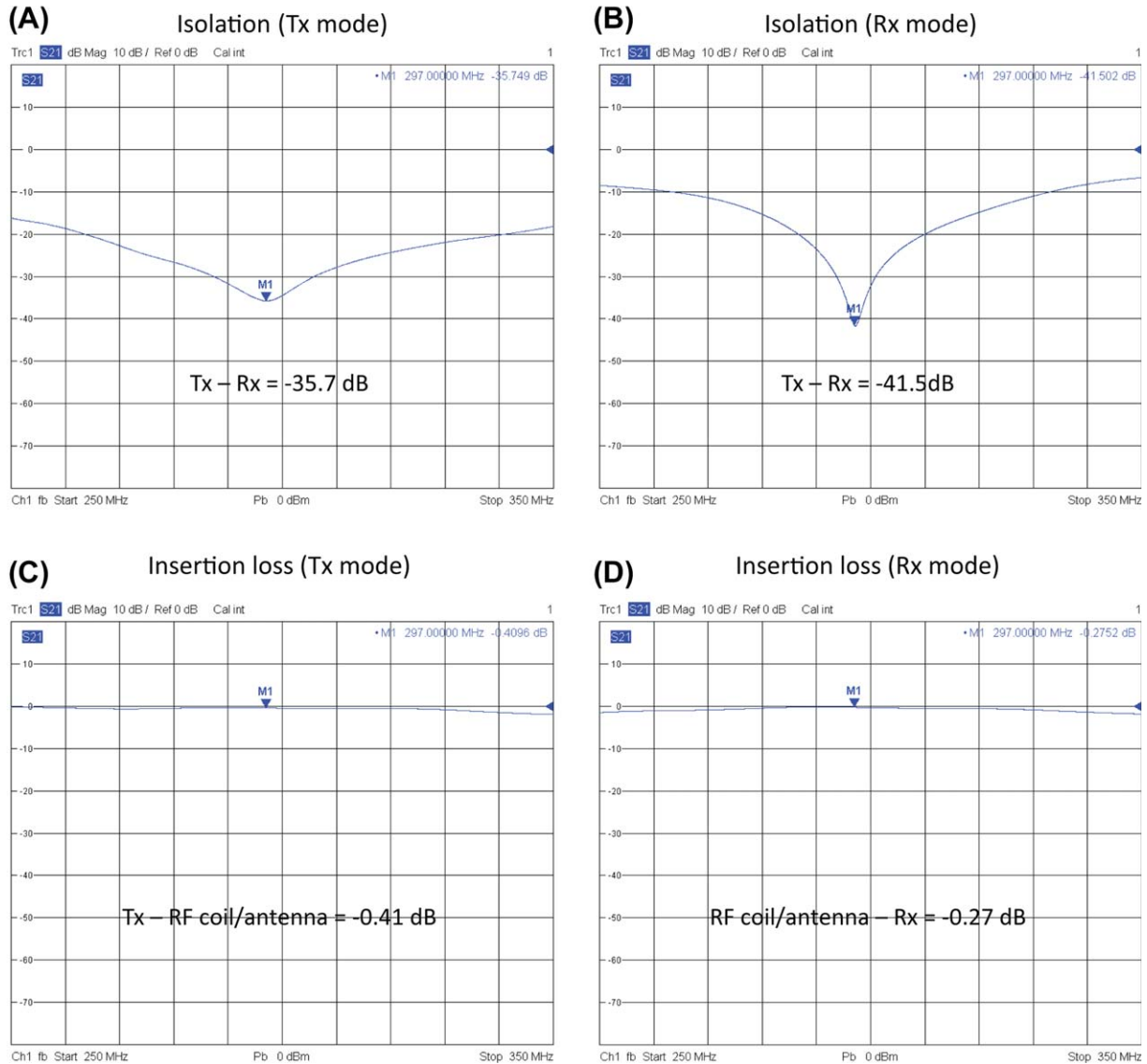


FIGURE 4 Network analyzer measurement results of the proposed high-power Tx/Rx switch showing isolation between Tx port and Rx port (-35.7 dB) during transmission (A), isolation between the Tx port and the Rx port (-41.5 dB) during reception (B), insertion loss between the Tx port and the RF coil/antenna (-0.41 dB) during transmission (C), and insertion loss between the RF coil/antenna and the Rx port (-0.27 dB) during reception (D)

For comparison, a conventional Tx/Rx switch was used, using the same heating paradigm with the exception that the average power was limited to $P_{\text{avg}} = 25$ W at the dipole antenna's feeding point.

2.5 | Temperature-induced release of a model drug

To demonstrate a potential application of thermal MR, the experimental setup for RF heating was used for thermally controlled release of molecules from temperature-responsive carriers. For this purpose, a thermoresponsive polymer P(DEGMA-co-OEGMA-*b*-[TMSPMA-co-VBA])²⁹ was loaded with fluoresceinamine, a fluorescent dye that served

as a model for a drug. In this study, the lower critical solution temperature⁷ of the thermoresponsive polymer is 34°C ; at a temperature greater than 34°C , the polymers will contract and become insoluble. This contraction releases the molecules trapped within the polymer layer into the bulk solution.

Solution containing fluoresceinamine-loaded polymer was transferred into sample holders of the phantom (Figure 3B): (1) 9 mL of the solution (hereafter referred to as the heated sample) was placed in the center of the hotspot created by the bow-tie dipole antenna; and (2) 9 mL of the solution was placed in a holder outside of the hotspot (the control sample). The distance from the dipole antenna to the phantom increased due to insertion of a tube (diameter = 3 mm) that

Power test

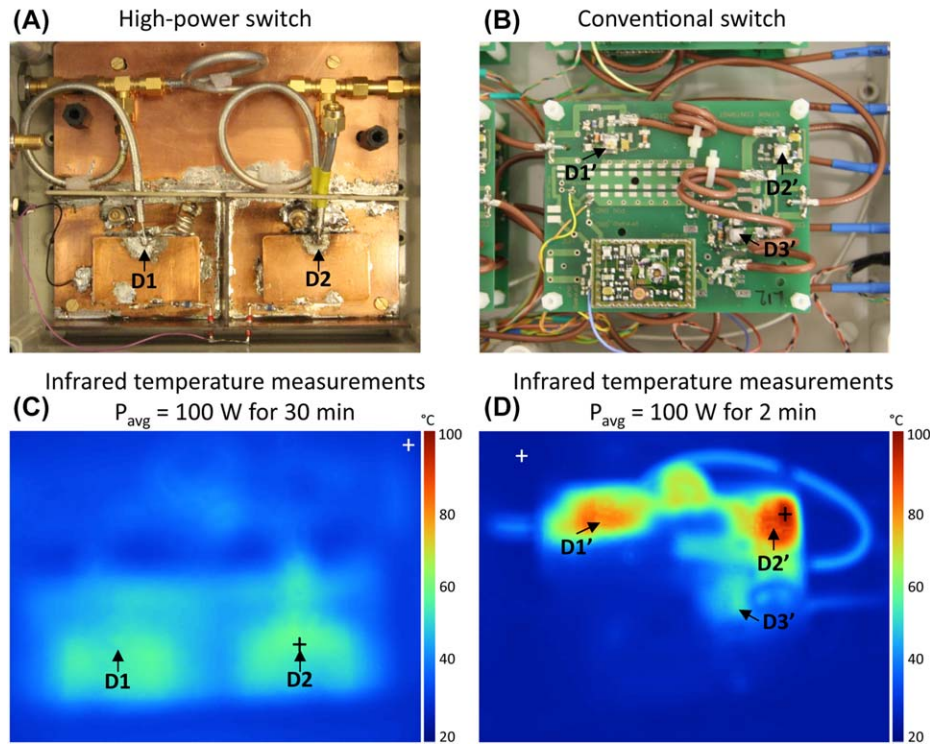


FIGURE 5 A, Photograph of the high-power Tx/Rx switch circuit. Black arrows point to PIN diodes D1 and D2. B, Photograph of the conventional Tx/Rx switch circuit. Black arrows point to PIN diodes; D1' and D2' are in the transmission path; D3' is in the reception path. C, Temperature map acquired with an infrared camera in the high-power Tx/Rx switch after 30 minutes of operating at $P_{avg} = 100$ W. Background temperature (white +) = 28.1 °C and highest temperature (black +) = 58.4 °C (PIN diode D2). D, Temperature map acquired with an infrared camera in the conventional Tx/Rx switch after only 2 minutes of operating at $P_{avg} = 100$ W. Background temperature (white +) = 24.9 °C. The highest temperature (black +) of 97.5 °C was found around the PIN diode D2'

would be used to extract the polymer sample. The base temperature in the phantom was maintained at 30 °C by a circulating water bath (Thermo Fisher Scientific, Waltham, MA).

Radiofrequency heating of the polymer-fluoresceinamine solution was performed with $P_{peak} = 1$ kW and $P_{avg} = 100$ W (4-ms rect-pulse, TR = 40 ms, duty cycle = 10%) at the antenna feeding point. After 30 minutes, the temperature in

In vivo wrist imaging at 7.0 T

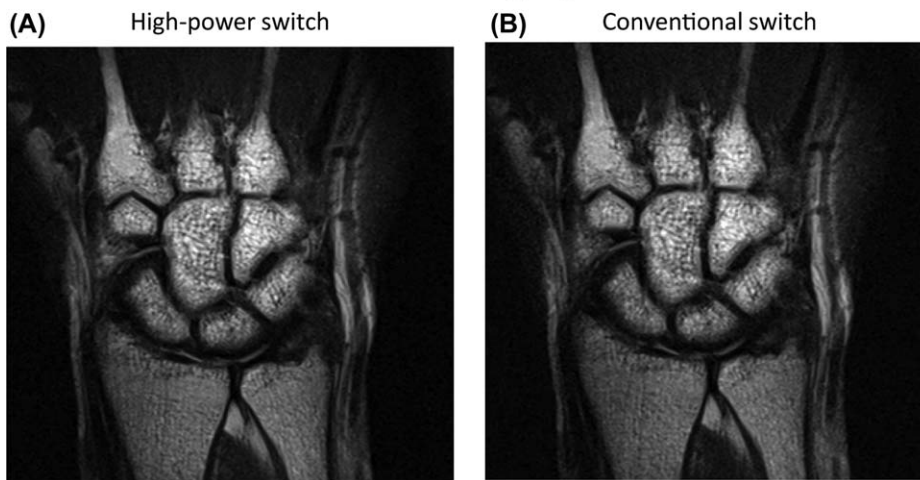


FIGURE 6 In vivo wrist MRI at 7 T using a RARE technique: in-plane resolution = 0.3×0.3 mm², slice thickness = 3 mm, matrix size = 320 × 320 mm², number of slices = 11, TR = 3000 ms, TE = 59 ms, nominal flip angle = 150°, $P_{peak} = 223$ W, $P_{avg} = 10$ W, total scan time = 149 seconds. A, Acquired with the high-power Tx/Rx switch. B, Acquired with the conventional Tx/Rx switch

the heated sample achieved 40°C, whereas the temperature in the control sample remained at 30°C. The heated sample was maintained at 40°C for an additional 120 minutes. During this time, small quantities of both heated and control samples were extracted and passed through size exclusion (6 kDa) chromatography columns (Bio-Spin 6, Bio-Rad Laboratories, Hercules, CA). In our case, the released fluoresceinamine (347 Da) would be retained in the gel matrix, whereas the fluoresceinamine still bonded to polymers (15 kDa) would be found in the eluate. The eluates were collected and analyzed in a fluorescence reader (GeminiXPS, Molecular Devices, San Jose, CA).

3 | RESULTS

3.1 | Transmit/receive switch performance

The proposed high-power Tx/Rx switch provided an isolation of -35.7 dB (Figure 4A) during transmission and -41.5 dB (Figure 4B) during reception. The insertion loss was -0.41 dB (Figure 4C) during transmission (Tx port – RF coil/antenna) and -0.27 dB (Figure 4D) during reception (RF coil/antenna – Rx port). The switching time was approximately 50 μ s.

After operating at an average power $P_{\text{avg}} = 100$ W for 30 minutes, the maximum temperature measured in the circuit of the high-power Tx/Rx switch was 58.4°C in the PIN diode D2 (Figure 5A,C). Applying the same average power, the conventional Tx/Rx switch could only be monitored for a period of 2 minutes, as the temperature in the circuit was already reaching critical levels (97.5°C close to the PIN diode D2' placed in the transmission path) (Figure 5B,D).

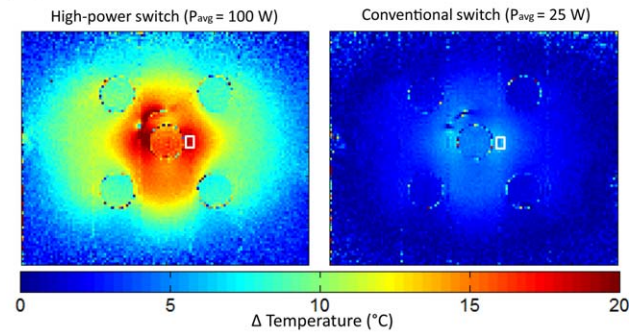
3.2 | Magnetic resonance imaging

The high spatial resolution MR images acquired with the high-power Tx/Rx switch (Figure 6A) and a conventional Tx/Rx switch (Figure 6B) yielded similar image quality. On both images, small structures such as the carpal bones, ulnar disc, radio-carpal joints, ulnar-carpal joints, and cartilage could be identified. The SNR was 30.3 for the high-power Tx/Rx switch and 25.4 for the conventional switch. In addition to the MRI experiments, the high-power Tx/Rx switch was tested successfully with the maximum available peak power of the MR system ($P_{\text{peak_max}} = 3930$ W at the switch).

3.3 | Radiofrequency-induced heating

With the applied heating paradigm, the temperature increase in the phantom, as observed by MRTh (Figure 7A), was $\Delta T = 5.9^\circ\text{C}$ after 5 minutes, 13.6°C after 10 minutes, and 18.5°C after 15 minutes. The fiber optic probe positioned at the same location registered $\Delta T = 6.7^\circ\text{C}$ after 5 minutes,

(A) MRTh temperature maps after 15 min of RF heating



(B) RF heating induced temperature changes

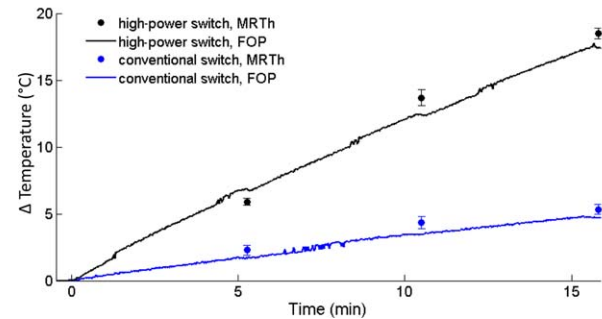


FIGURE 7 A, Temperature maps obtained from MR thermometry (MRTh) (in-plane resolution = 1.5×1.5 mm², slice thickness = 4 mm, matrix size = 192×192 , number of slices = 3, TR = 61 ms, TE1 = 2.26 ms and TE2 = 11.44 ms, nominal flip angle = 30° , total scan time = 11 seconds) after 15 minutes of RF heating with the high-power Tx/Rx switch and $P_{\text{avg}} = 100$ W at the antenna feeding point (left) and with the conventional Tx/Rx switch and $P_{\text{avg}} = 25$ W at the antenna feeding point (right). B, Radiofrequency heating–induced temperature changes at a depth of 15 mm in the phantom monitored with MR thermometry (evaluated in the white rectangle in [B]) and fiber optic probes (FOPs)

12.4°C after 10 minutes, and 17.5°C after 15 minutes (Figure 7B).

In comparison, the maximum temperature increase in the same location while using a conventional Tx/Rx switch at its limit of $P_{\text{avg}} = 25$ W was only $\Delta T = 2.3^\circ\text{C}$ after 5 minutes, 4.3°C after 10 minutes, and 5.3°C after 15 minutes, as determined by the MRTh temperature map, and $\Delta T = 1.7^\circ\text{C}$ after 5 minutes, 3.5°C after 10 minutes, and 4.7°C after 15 minutes registered by the fiber optic probe.

3.4 | Temperature-induced release of a model drug

The amount of fluoresceinamine released from the thermoresponsive polymers in a heated sample versus a control sample is shown in Figure 8. The nonheated control sample released only a small amount of fluoresceinamine over a timeframe of 120 minutes. This is due to the slow hydrolysis of Schiff base bonds that attach fluoresceinamine to the polymer at low temperature. The heated sample subject to a temperature above lower critical solution temperature of the polymer

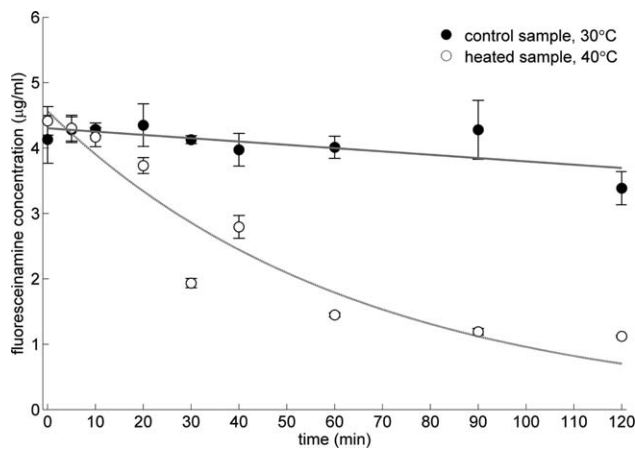


FIGURE 8 Release of fluoresceinamine from the thermoresponsive polymers after RF heating. After reaching 40 °C in the heated sample, the release of fluoresceinamine was monitored over 120 minutes. The control sample, maintained at 30 °C (greater than the lower critical solution temperature of 34 °C), showed a very slow release. This is primarily due to slow hydrolysis of Schiff base bonds that attach fluoresceinamine to the polymer. The heated sample, subject to a higher temperature and above the lower critical solution temperature of the polymer, showed a higher release rate: 68% after 60 minutes. In this case, the Schiff base bond was hydrolyzed faster at higher temperatures, and the contraction of the polymer upon the temperature being greater than the lower critical solution temperature increased the release of fluoresceinamine

(lower critical solution temperature = 34 °C) showed a higher release rate (e.g., 68% after 60 minutes). In this case, the Schiff base bond was hydrolyzed faster at higher temperatures, and contraction of the polymer at temperatures greater than the lower critical solution temperature increased the release of fluoresceinamine.

4 | DISCUSSION AND CONCLUSION

This work adds to the literature by proposing a high-power Tx/Rx switch that is able to support thermal MR applications of high peak power MRI and high average power RF heating. In our experimental implementation, the high-power switch was able to withstand an average power of $P_{\text{avg}} = 120$ W and a maximum peak power of $P_{\text{peak}} = 3.9$ kW. The proposed configuration afforded a low insertion loss of -0.41 dB/ -0.27 dB (transmission/reception) and sufficient isolation of -35.7 dB/ -41.5 dB (transmission/reception), similar to a commercially available Tx/Rx switch with a measured insertion loss of -0.55 dB/ -0.32 dB (transmission/reception) and isolation of -35.6 dB/ -46.8 dB (transmission/reception). With 50 μ s, the measured switching time for the high-power switch was slightly lower than for the conventional switch (approximately 100 μ s).

The proposed design is simple and fairly low cost. It consists of nonmagnetic components, allowing it to be placed

inside the scanner bore in close proximity to the RF antenna. The strategic placement of the $\lambda/4$ stubs routes the RF signal from the transmitter directly to the RF coil/antenna without directly passing through any lossy electronic components such as PIN diodes.²⁷ This enables the use of high average powers in comparison to other Tx/Rx switches designed for MRI applications.^{22–24} For instance, the conventional Tx/Rx design that was used as a reference in this study uses PIN diodes in the transmission path that exhibit a series resistance of 0.35 Ω and 2 pF. The thermal resistance of these PIN diodes is 5.5 °C/W and the junction temperature is 150 °C. For an ambient temperature of 20 °C, these PIN diodes could only withstand approximately 24 W. Stud mounting, heat sinks, and lower resistance PIN diodes might improve the maximum average power handling of this circuit; however, it will not be sufficient to increase the applicable power by a factor of more than 4 to above 100W, which is the applicable power of the proposed switch design in this study. The implementation of air or water cooling circuits that provide sufficient cooling to an array of the proposed high-power switches is cumbersome if not prohibitive for human studies in a long-bore 7 T MR scanner.

In L-band radar applications, GaN-HEMT Tx/Rx switches have been presented that facilitate high-power (200 W) handling.³⁰ However, the high breakdown voltage requirements for high peak power MR pulses will necessitate the use of multiple transistors per switch, which increases the insertion loss, and in particular the costs per switch, significantly. These high costs are not beneficial for the setup of a multichannel (e.g., 16-channel or 32-channel) transmission system that is needed for thermal MR.

The length of $\lambda/4$ stubs is 15.5 cm at 7 T ($f = 297$ MHz). At higher fields, which are in particular interesting for thermal MR applications,¹⁹ the size of the $\lambda/4$ stubs would be reduced, allowing for a more compact switch design. For instance, at $B_0 = 10.5$ T ($f = 447$ MHz), the $\lambda/4$ stub size is reduced by approximately 33%.

Using microstrips instead of semirigid cables could be another approach to reduce the current size of the high-power Tx/Rx switch, which could increase transmission losses of the circuit.^{27,31}

In reception mode, ground is well defined by the low noise preamplifier; however, in transmission mode this might not be the case due to the long looped quarter wave coaxial cables used. Thus, the frequency-selective ground at location B (Figure 1A) is crucial to improving isolation. At the same time, interferences with other RF frequencies are reduced. This is of importance for RF heating at higher RF frequencies (e.g., 600 MHz) in a hybrid system in which MRI and/or RF heating are performed simultaneously at 7 T.^{19,32} The frequency-selective ground might potentially be affected by the coax cable connecting the antenna to the antenna port of the switch through coupling effects, which might degrade its

isolation performance. To reduce the influence of the connected coax cable on the switch, either a cable trap at the antenna feeding cable or a trap circuit incorporated into the Tx/Rx switch design at the RF coil/antenna port can be implemented.

High-resolution MRI of the wrist was supported by the high-power switch, with comparable image quality benchmarked against a commercially available Tx/Rx switch. Radiofrequency heating ($\Delta T > 16^\circ\text{C}$) and MRTh were performed using the high-power switch. The MRTh results are well correlated to the readings obtained from fiber optic probes, demonstrating the feasibility of this application without the use of additional cooling circuits for the Tx/Rx switch. These findings allowed the use of the switch for thermally controlled release of a model drug from thermoresponsive polymers in a phantom.

In conclusion, the proposed high-power Tx/Rx switch enables thermal MR applications at 7 T, contributing to the study of the role of temperature in biological systems and diseases with possible implications for diagnosis, treatment, and MR safety. To facilitate reproducibility and swift dissemination, all design files of the switch will be made available open-source at www.opensourceimaging.org.

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CONFLICTS OF INTEREST

Thoralf Niendorf is the founder and CEO of MRI.TOOLS GmbH. Helmar Waiczies is an employee of MRI.TOOLS GmbH.

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